FAST National University of Computer and Emerging Sciences

H-11/4, Islamabad

CS3014: Applied Human Computer Interaction
D1A: Research Questions, Materials and Protocol

Team Name: MSM Technologies **Section:** G **Stage:** Empathize (Weeks 4-5)
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Part 1: Research Questions

Our D0 Problem-Space Brief made two explicit, unverified assumptions about the FAST-NU Insignia library portal (nu.insigniails.com/library). This deliverable turns those assumptions into questions about what students already do, not about what the portal should become.

1.1 Primary Research Question (RQ1)

Traces to: D0 Assumption 1 (Room Booking)

RQ1: How do FAST-NU students currently go about securing a library study room, and at which point in that process (locating the booking feature, completing the date and duration entry, or confirming the booking) do they stall, make errors, or abandon the portal in favour of the front desk?

D0 assumed the process is frustrating both in navigation (My Account, then Booking, then Book Room) and in execution (date picker and duration slider), while recording that the team does not know which of the two is the real drop-off point, or whether students use the portal at all. RQ1 is scoped to settle exactly that. It treats booking in person as a current behaviour to be documented rather than a failure to be corrected.

1.2 Secondary Research Question (RQ2)

Traces to: D0 Assumption 2 (Book Search)

RQ2: When a FAST-NU student needs a specific book, what sequence of steps and sources do they actually use to locate it, and how do they decide between, or avoid, the portal's parallel search modes?

D0 assumed the four undifferentiated modes (Simple, Advanced, Kiosk, Assets) confuse students, that most default to Simple search, and that few know Advanced search filters by author, subject or call number. It also recorded the competing possibility that students skip portal search entirely. RQ2 is framed around the whole book-finding journey rather than the portal screen, so a bypass route shows up in the data instead of being screened out by the question.

1.3 Sub-Questions and Traceability

These are not additional research questions. They are the specific unknowns each RQ breaks into, and each maps onto a protocol section in Part 4.

Ref	Sub-question	Resolves	Protocol
RQ1a	How do students first learn that room booking exists, and how do they re-find it later?	Discovery or recall	4.3 Segments B, C

Ref	Sub-question	Resolves	Protocol
RQ1b	What errors occur during date and duration entry, and how do students notice and recover?	Whether the controls produce wrong bookings	4.3 Segment C
RQ1c	When do students book at the front desk instead, and what makes that route preferable?	Whether the portal is bypassed, and why	4.3 Segments B, E
RQ2a	Where does a book search begin: portal, Google, a classmate, staff, or the shelf?	The real starting point	4.3 Segments D, E
RQ2b	What do students do when results are too many, too few, or irrelevant?	Whether filtering is attempted at all	4.3 Segment D
RQ2c	How do students read the four search modes when they meet them unprompted?	Whether mode choice is a real decision	4.3 Segment D

1.4 Why These Questions Are Not Solution-Oriented

Neither question names a feature, a fix or a redesign. A plausible answer to either is that no problem exists, for example that students book rooms comfortably in two clicks, or that the four modes are never noticed because Simple search is the landing default. The protocol is written so those outcomes would be visible in the data rather than argued away.

Part 2: Participant Criteria and Sampling Strategy

2.1 Inclusion Criteria

A participant is eligible if all of the following hold:

- Currently enrolled at FAST-NUCES Islamabad as an undergraduate or postgraduate student, with an active library account.
- Aged 18 or over, confirmed at screening.
- Has needed a study room, or has needed to find a specific book, at least once in the current or preceding semester, whether or not the portal was used.
- Willing to attempt one live task on their own phone or laptop while thinking aloud, in a session of roughly 25 minutes.

The third criterion is written around the need rather than around portal usage. Screening on portal usage would silently exclude the students who abandon or bypass the system, which is precisely the group D0 flagged as unverified.

2.2 Exclusion Criteria

- Anyone under 18. This is a hard exclusion, so the study collects no data from minors and needs no formal ethical review on that ground.
- Members of MSM Technologies and of Team HAT, who piloted our protocol. Both groups know the study's assumptions and cannot give uncontaminated evidence.
- Library staff, administrators and IT personnel. They are secondary and tertiary stakeholders in our D0 map and would require a separate protocol and separate consent.
- Students who have not visited the library in the current or preceding semester, and anyone unwilling to have written notes taken, since no usable record could be produced.

2.3 Target Sample Composition

Segment	Target n	Rationale
Undergraduates, years 1 and 2	3	Newer to the portal, so discovery and first-run problems surface (RQ1a)
Undergraduates, years 3 and 4	3	Established habits and workarounds (RQ1c)
Postgraduate students	2	Different search needs: author, subject and call-number lookups (RQ2)
Front-desk bookers, any year	2, may overlap the above	Direct evidence on whether the portal is bypassed rather than merely disliked
Total	8 to 10	Enough for behavioural saturation at this scale and feasible in the time available

2.4 Sampling Strategy

Named strategy: purposive sampling, with a snowball component for the postgraduate and front-desk segments.

Purposive is correct here because the criteria are specific. We are not sampling students in general but students who have recently faced the two tasks in RQ1 and RQ2, split across the segments above, and a screener enforces this. Convenience sampling in the CS labs would over-represent third-year Computer Science students who share our own habits, and would likely miss postgraduates and non-portal users, leaving both D0 assumptions untested where they are weakest.

Snowball is added because those two segments cannot be reached by screening a corridor. Postgraduates are a small, dispersed population, and front-desk bookers are invisible until someone identifies them, so each participant is asked to refer one classmate matching a segment we still need. Recruitment is still limited to students we can reach on campus within a two-week window, which we record as a stated limitation in D1B rather than claiming a representativeness we cannot support.

2.5 Screener and Recruitment

Recruitment is face to face in the library, the cafeteria and departmental corridors, plus referrals. Each candidate answers four questions before any consent is taken:

1. Are you enrolled at FAST-NU Islamabad, and are you 18 or older? (Yes to both is required.)
2. Which programme and year are you in? (Assigns segment.)
3. This semester or last, have you needed a study room, and if so how did you end up getting one? (Open; assigns portal user or front-desk booker.)
4. In the same period, have you needed to find a specific book? (Yes to question 3 or 4 is required.)

Screening answers are recorded against a participant ID only. If a candidate is ineligible, the note is destroyed on the spot.

Part 3: Consent Materials

3.1 Consent Script

Read aloud, verbatim, by the facilitator before any note-taking begins. A printed copy is offered to the participant to read along.

> Hello, my name is Muhammad Umar and I am a student in CS3014, Applied Human Computer Interaction, at FAST-NU Islamabad. My team is studying how students use the university library portal to book study rooms and to find books. We are studying the system, not you, so there are no right or wrong answers here.
>
> The session takes about 25 minutes. I will ask how you normally get a study room and how you normally find a book, and then ask you to try one of those tasks on your own phone or laptop while saying out loud what you are doing.

A teammate sitting with us will take handwritten notes, and may write down some of your exact words in quotation marks so that we report them accurately. Nothing is audio or video recorded. We do not record your screen, photograph you, or note anything from your library account.

Your name and roll number are not written down anywhere. You will be referred to only by a code such as P04. The notes are kept in a folder only my four team members can open, are shown to our course instructor on request, and are deleted at the end of this semester. We will not upload anything that could identify you to any external AI tool or online service.

Taking part is voluntary and has no effect on your grades, your library account or your standing in the university. You can skip any question or stop the session at any point without giving a reason, and if you stop, I will destroy the notes taken so far.

Do you have any questions before we begin? [Pause for questions.]

Do you agree to take part in this session? [Record yes or no in the consent log.]

Note on consent format

Consent is taken verbally and logged rather than by signature. A signature sheet would create the only identifiable record in the study and would then have to be stored and protected separately, whereas verbal consent against a participant ID serves the same evidentiary purpose without collecting a name. Since no audio or video is collected, only one consent decision is asked of the participant.

3.2 Consent Log (one row per session)

Participant ID	Date	Segment	Consent to participate	Facilitator initials
P01				
P02				
…				

3.3 Data-Management Plan

Item	Plan
What is collected	Handwritten notes, including short verbatim phrases in quotation marks, and one structured observation sheet per session (4.4). No audio, no video, no screen recording, no screenshots, no names, no roll numbers, no photographs.
Identifiability	Participants are assigned an ID (P01 to P10) at screening, and the ID is the only identifier on any artefact. No key linking IDs to names is created at any point, so the dataset is anonymous from the moment of collection.
Storage and access	Paper notes stay with the note-taker until they are typed up within 24 hours, into one access-restricted folder on the team's university Google Drive with link sharing disabled. A second member checks the typed version against the paper original, after which the paper is shredded. Access is limited to the four named members of MSM Technologies, plus the course instructor or coordinator on request.
Anonymisation	Nothing identifiable is recorded, so anonymisation is enforced at capture. While typing up, the transcriber replaces any identifier a participant happened to speak aloud, such as a person's name or a staff member's name, with a bracketed placeholder like [name].
Deletion	All raw data, meaning typed notes, observation sheets, the consent log and screening notes, is deleted from the shared folder and all local copies within two weeks of the final project deliverable being graded. Only aggregated, anonymous findings survive in D1B and later deliverables.

Item	Plan
External tools	No session data is uploaded to any external AI tool or third-party service. Where an AI tool is used at all, it is used only on text the team has already written, such as checking the phrasing of this protocol, never on participant data.

3.4 Hard Ethical Constraints: Confirmation

- **No vulnerable populations.** Participation is limited to enrolled students aged 18 or over, verified at screening. No minors, patients or other vulnerable groups are involved, so no formal ethical review is triggered.
- **No identifiable data to external AI tools.** The study collects no identifiable data at all, and no session data is uploaded to external AI tools. Participants are told this directly in the consent script.
- **Voluntary throughout, and no deception.** Consent is taken before anything is written down, withdrawal is available at any point without justification and destroys that participant's notes, and participants are told at the outset that the study concerns the library portal's usability.

Part 4: Piloted Protocol

This is a semi-structured interview combined with a short observed task session using concurrent think-aloud. The version below is the revised, post-pilot version, and is the one we intend to field in D1B.

4.1 Pilot Details

Item	Detail
Piloted with	Team HAT, a fellow CS3014 team, whose members took the participant role
Format	One full run of the draft protocol, followed by a debrief in which the pilot participants critiqued the questions themselves
Duration	The draft run overran the intended session length; the revised protocol targets 25 minutes
Observed by	Shahmeer Atif, recording timing and question-by-question reactions
Outcome	Three changes to the instrument, listed in 4.5; the version in 4.3 is the revised one

4.2 Session Structure

Seg.	Segment	Time	Purpose
A	Consent and framing	3 min	Read consent script, record consent, set expectations
B	Study room, current behaviour	5 min	RQ1a and RQ1c, with the portal not yet mentioned
C	Study room, observed attempt	7 min	RQ1a and RQ1b, where the process actually breaks
D	Finding a book, observed attempt	7 min	RQ2a to RQ2c, the real search journey and mode choice
E	Workarounds and closing	3 min	RQ1c and RQ2a, bypass routes and referrals

4.3 Protocol: Questions as Asked

Segment A: Consent and framing (3 min)

- Read the consent script in 3.1 verbatim. Record consent in the log before anything is written down.
- "Through this session, please keep saying out loud what you are looking at and what you are trying to do. If you go quiet I will just nudge you with 'what are you thinking now?', that is all."

Segment B: Study room, current behaviour (5 min)

The portal is not named in this segment. The first mention of it must come from the participant.

1. Think about the last time you needed a quiet place to work on campus with a group. Walk me through what you did, from the moment you realised you needed the space.
2. How did that turn out? Did you get the room you wanted, and is that how it usually goes?
3. How did you first find out that was the way to do it?
4. [Only if the portal has been mentioned] The last time you booked through the portal, how did you get to the booking page? Show me on your phone if that is easier.
5. [Only if the portal has not been mentioned] Have you ever booked a room any other way?

Segment C: Study room, observed attempt (7 min)

The participant uses their own device and account, and is asked to stop before final confirmation so that no real booking is created.

1. "Start from wherever you would normally start, and get as far as booking a room for two hours tomorrow afternoon. Stop just before you would confirm it, do not actually submit. Keep talking as you go."
2. [Silence probe] "What are you looking for right now?"
3. [After a hesitation or backtrack] "You paused there. What were you expecting to happen?"
4. [After completion or abandonment] "How did that compare to what you expected?"
5. [If they abandoned] "At what point would you normally have given up and done something else?"

Segment D: Finding a book, observed attempt (7 min)

1. "Is there a book you actually need at the moment, for a course or anything else?" [If yes, use it. If not, use the fallback: "Suppose your instructor has told you to read Sommerville's Software Engineering this week. Find out whether the library has a copy you can borrow."]
2. "Start wherever you would normally start. That does not have to be the library site. Keep talking as you go."
3. [If they use portal search] "How did you decide to search that way?"
4. [If results are too many or unhelpful] "What would you do next?"
5. [Only if the participant opens or names a second search mode] "What do you think that one does differently?"
6. "If this had not worked, what would you have done instead?"

Segment E: Workarounds and closing (3 min)

1. Is there anything about the library, whether rooms, books or fines, that you have worked out a shortcut for that a first-year would not know?
2. Is there anything you have simply stopped trying to do on the portal?
3. Is there anything I should have asked you about that I did not?
4. [Snowball] We are still looking to speak to [postgraduate students or people who book rooms at the desk]. Could you pass our contact to someone? They would contact us, not the other way round.
5. Thank the participant, confirm what happens to their notes, and confirm that their participant ID is the only record.

4.4 Observation Recording Sheet

Completed by the note-taker during Segments C and D while the facilitator runs the session, one sheet per participant, filed under the participant ID.

Field	What to record
Navigation path	Every click or tap in order, from starting point to end state
Time to locate	Seconds from task start to reaching the booking page or the results page
Errors	Wrong date, wrong duration, wrong mode opened or wrong item selected, and whether the participant noticed

Field	What to record
Recovery and hesitation	What they did after each error, whether they recovered unaided or abandoned, and any pause of roughly three seconds or longer with what was on screen at the time
Verbalisations	Exact wording where the participant states an expectation, a confusion or a decision rule
End state	Completed, completed incorrectly, abandoned, or switched to a non-portal route
Facilitator interventions	Any prompt given beyond the scripted probes, flagged because it contaminates that data point

4.5 What the Pilot Changed

The draft protocol was run against Team HAT, whose members took the participant role and then critiqued the instrument directly. Three changes came out of it, all affecting the evidence the protocol can produce.

#	Before pilot	After pilot	Why it was changed
1	Segment B opened by naming the library portal and asking how often the participant used it.	Segment B now opens with the need itself, a quiet place to work with a group, and never names the portal first.	Naming the portal steered the conversation onto it. The debrief made clear that participants would describe portal use because we had put it in front of them, hiding any student who books at the front desk instead, which is the exact possibility D0 left open.
2	"What did you find most frustrating about the room booking slider?"	"Walk me through what happened the last time you booked a room." Frustration is followed up only if the participant raises it.	The question carried our own D0 conclusion inside it and invited agreement rather than information. Any answer would have supported Assumption 1 whether or not the assumption is true.
3	Six demographic questions at the start: programme, year, CGPA, hostel or day scholar, device, library visit frequency.	Reduced to programme and year, both already captured in the screener, so the session now begins with Segment B.	They consumed several minutes, contributed nothing to RQ1 or RQ2, and were the main reason the pilot run overran.

The protocol in 4.3 is the version these changes produced, and is what we intend to field in D1B.

Part 5: Submission Checklist

#	Requirement	Where addressed
1	Research question(s) stated and traced to a named D0 assumption, not solution-oriented	1.1, 1.2, 1.3, 1.4
2	Participant criteria and a named, justified sampling strategy	2.1 to 2.5
3	Consent script and data-management plan complete	3.1, 3.3
4	Piloted, revised protocol with a note on what changed after piloting	4.1, 4.3, 4.5
5	Hard ethical constraints confirmed	2.2, 3.4

Square-bracketed items in the consent script and the protocol are instructions to the facilitator. They are conditions and prompts to act on, not lines to read aloud to the participant.